Foundation for Virtual and Augmented Reality Systems

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Week – 05

Lecture – 24

Lec-24: XR as Interactive Systems

Hello and welcome to the NPTEL MOOCs course on Foundation for Virtual and Augmented Reality Systems. We have started our discussion on a new topic, that is Interactive Systems Usability and Extended Reality, as part of new lecture topic number 12. So, we will continue our discussion on this topic in this lecture as well. So, earlier we talked about the idea of interactive systems. These are systems to be used by laypersons who need not be technology experts, who need not have knowledge of the technological jargon to understand the system, to understand how it works.

We also touched upon the key considerations in building such systems, that is, what are the key challenges and what are the crucial things that we should be aware of while trying to build such systems. So, we will continue our discussion in this lecture. Now, earlier we talked about the importance of knowing the users. Typically, to understand who the users are and how they behave, we require some studies, some empirical studies or field studies to observe their behavior or to understand how they behave in a given situation. given context and so on. That is time-consuming, that is quite laborious, requires elaborate setup, elaborate planning and so on. However, it is not necessary to always start by studying the users in a field setting, in a practical setting. Instead, we can rely on well-known user categorization.

One such categorization is A three-class classification of users: novice, that is the first class; then intermittent, second class; and then expert, that is the third class of these categories. So, these three classes, namely novice, intermittent, and expert, can give us some starting point in trying to understand the users. Users that belong to each of these categories are likely to behave differently as compared to users in the other categories. And

broadly, we have some knowledge; we have some understanding of how users in each of these categories behave at a very broad level.

With that knowledge, certainly we can take design decisions while building interactive systems if we assume that our potential users will be of these three classes or any of these classes. Let us quickly go through one interesting example. All of you might have used some word processing system such as MS Word. Now, if you have noticed, there are different ways to perform the same task in any word processing system. Let us take the example of MS Word. It is equally applicable for other processing applications as well. Suppose you have created a file. Now you want to save it.

How can you save it? There are multiple options provided. One option is the combination of hotkeys. That is, you press the CTRL and S keys together on the keyboard to save the file. That is a hotkey combination.

There is another way of saving it. That is, you can use the file menu option. Under this, you can use the save option. Then a dialog appears. You provide the details and you can save it. So, through the menu option also you can save. Now, have you ever wondered why these multiple options are given when any one option is good enough to do the job? The answer lies in what we have just discussed, that is, these options are given to cater to different groups of users. Typically, when we are talking about novice users, they are basically the users who are using it for the first time.

For novice users, remembering hotkey combinations is generally considered difficult. So, they have to recall the knowledge. Because they are using it for the first time, they are unlikely to have created that knowledge itself. So, there is no question of recalling that this hotkey combination will do the job. So, this menu option is given through which they can save it.

Of course, that is, It takes a longer time as compared to the hotkey combination, but for novice users, it is suitable. For intermittent or expert users who have used the system for a longer duration or multiple times, the hotkey combination may provide a better alternative as compared to the menu option. Particularly for expert users, for them the hotkey combination is always preferable as compared to the menu option. Now, one thing to note here is that when we are talking about expert users, they are not really technology experts, rather they are experts in using the system.

So, they might have used the system for quite some time, already understood all aspects of the system, and they can use it very efficiently. So the three categories—novice, intermittent, and expert—are one way of classifying. Just to recap, novice means first-time usage, intermittent means occasional usage, and expert means regular usage over a long period of time. Of course, this is only one way of categorization, and there are more fine categorizations possible. Better categorizations can be done based on empirical studies of the users, but this broad categorization serves important purposes when we are starting our design process.

So, knowing the user is the first step. How do I consider the knowledge of what the users are expecting? what the user needs in the design. Let us have a little bit of discussion on that aspect. So, when we say we want to design a product or a system, there can be different design perspectives based on who the designer is.

The same term user-centric design can be viewed by different stakeholders in particular the designers in different ways. So, that is another important thing to note. If we are talking of a user interface or user experience designer, then such a designer is primarily concerned about the design of the interface elements and layouts, which is mostly a creative design aspect, informed by user characteristics. For an industrial product designer, it might refer to the form of the product that is shape, size and look. That is another kind of design activity. For an electronic engineer, a third type of designer, They might look at it as adding more features at the hardware level such as more sensors, smaller chips to reduce size and more power efficient battery, primarily from the hardware point of view.

Our main concern will be from the point of view of an application software developer, from the point of view of an application software developer. Because we will be dealing with software development. That is our primary concern. So for such a developer, the developer needs to follow a development lifecycle that takes care of the user characteristics to better match the user's needs and expectations. So when we are trying to build software, say for example, a VR system or an AR system, that is software, so we need to follow a particular development lifecycle or design lifecycle. In the lifecycle, we should be able to take care of users' characteristics and needs and expectations to make it easier for the user to use. So that is how we can take care of the user characteristics. by following an appropriate development lifecycle. Now, when we are following it, how do we know that we are actually able to take into account the user's characteristics?

This brings us to the important concept of usability. Usability tells us how good the system is from the point of view of the user. So, essentially what we require is an explicit measure a measure to judge our design if we have taken care of the human factors. Now, this measure is called usability.

So, essentially it is a measure of the goodness of a system from the point of view of the user. Like before, let us go through the definition of usability. So, we will be following the ISO, International Standards Organization definition. It is the ISO 9241-210-2009 standard definition of usability, which is the extent to which a product can be used by specified users to achieve specified goals. So, used by specified users to achieve specified goals.

with effectiveness, efficiency, and satisfaction in a specified context of use. So, these are crucial considerations when we are talking about building a usable product, a product that is supposed to have taken into account users' needs and expectations. So, there are three crucial aspects in this definition. First is the product is meant to be used by a specified group of users. In other words, a usable product need not be designed for all.

So, when we start building our system, we should not try to think that our system is going to be used by anybody and everybody whoever comes. Instead, we should try to clearly identify the target user groups and accordingly try to build a usable system. Only then is it likely to be able to satisfy the users. Otherwise, if we start by assuming that it is meant for everybody, then most likely it will not be able to satisfy all possible users. So, we should always try to identify the specified group of users for whom we are building the system.

The product should allow the users to achieve a specified set of goals. So, we should not try to let the user do anything. Instead, we should allow the users to perform specified tasks to achieve specified goals. So putting every conceivable feature in a product does not necessarily lead to a usable product. That is another crucial aspect of the definition.

That is, when we are building a system, we should not assume that it must be feature-rich. It should have all possible conceivable features as part of the package. In that case, users are most likely to get confused. They have specified goals. They have specified goals.

They have specified needs and expectations. So only those features we should carefully identify and support in our design. Thirdly, the product should be designed for a specified context of use. Clearly, a usable product need not be so for all usage scenarios. So, even if we are assuming that we have specified the users and we have specified features catering to the needs and expectations of the users.

Still, we should not expect that even for those users it should be usable for all usage contexts. Ideally, the usability is restricted to a specified context of use. So, these three are very crucial: we should not try to build systems for everybody, and accordingly we should not try to make it as feature-rich as possible, and thirdly we should not expect The specified users to be happy always whenever they are using the system. Instead, it is supposed to make them satisfied when they are using it in a specified context of use.

Also, the ISO definition of usability reveals three measures. Usability, as we have started with, is a measure of how good a system is from the point of view of the user. So there are three crucial aspects, but along with that, there are three measures of usability. What are those measures? First is effectiveness, how effective the product is; efficiency, how efficient it is; and satisfaction, whether the user is satisfied.

So, effectiveness essentially refers to whether the user is able to do whatever they want to do. Efficiency refers to how fast the user can do what they want to do, and satisfaction refers to how satisfied the user is when they are using the system. So, these are broadly the metrics to measure usability. Exact measurement techniques vary; it can be objective, it can be subjective, like we have seen before for measurement of presence or cybersickness, etc. So, with this knowledge, we are now ready to introduce the term user-centered design.

So, earlier we talked about user-centric approach. And using that approach, whatever design comes up, that is called user-centered design. Now, this term was coined by Ben Shneiderman way back in 1986 in the context of graphical user interfaces that we are so familiar with. The objective was to design products that increase usability. Now when we are talking of user-centered design, it actually indicates involvement of the users either in an active manner or in a passive manner in the design lifecycle itself.

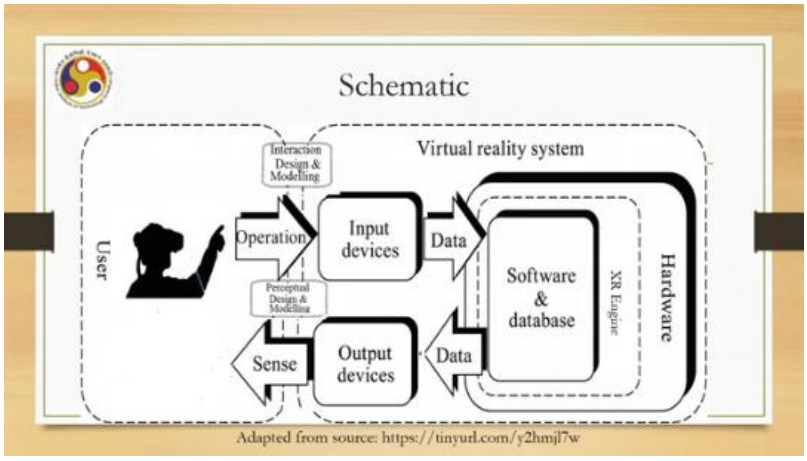
So, remember that we talked about having a design or development lifecycle when we are building a software application, a usable software application. Now, when we are talking of user-centered design, what it refers to is involving users in that lifecycle itself either actively or in a passive manner. There are a few related terms which refer to a more or less similar idea. Cooperative design coined by Greenbaum and Kyng in 1992. Participatory design, Schuler and Namioka, 1993, contextual design, Beyer and Holtzblatt, 1997.

So, these terms are more or less similar in concept as compared to the user-centered design term. ISO also has its own standard term So, it prefers to use the term human-centered design rather than user-centered design. So, all of these are referring to the same concept, that is to take users into the design process. So, in user-centric software design, software

development, the core design concern is the design of usable systems to cater to the needs and expectations of the lay persons or layman users. So, we now have got some understanding of what we are talking about. So, we talked about interactive systems that are systems to be used by lay persons who need not understand the technology behind. Now for them, we need to know the users, who the users are. There is a broad categorization of users we talked about.

That is novice, intermittent, and expert. Novice are those who are using it for the first time, intermittent are those who are occasional users of the system, and experts are those who have used it regularly for a longer period of time. Now, these users need to be actively or passively involved in the overall software development lifecycle that we are supposed to choose to build our system, that is the user-centered software development approach. Now, that is about interactive systems.

Let us try to relate this with the idea we have, that is extended reality systems. Now when we are talking of extended reality software, what is our core concern? That is to create a relationship between the user and the virtual environment. Now here, of course, we know that both software and hardware play an important role in building the relationship. Now if I recollect the relationship between the user and, say, a virtual reality system.



We have the user at one end; the user uses the input devices that we have discussed earlier, that is indicated by this operation part. Now, interaction with the input devices generates some input data that is sent to the system. The system comprises hardware, the middleware that is the XR engine, and the application layer. The data gets processed here in the system. Output data is generated which is sent to the output devices.

Output devices stimulate our senses, that is, the user senses. Now, while providing input data we should assist the user to provide input data seamlessly, effectively, easily following a user-centered design approach which is captured here interaction design and modeling how to provide input from the user's point of view how easily to provide input. Similarly, how well the output is generated is part of perceptual design and modeling that is also crucial because, if you can remember, we have talked about generating stimuli, generating appropriate stimuli and appropriately perceiving the stimuli are both important. So, in order to ensure that these two match, we must take recourse to perceptual design and modeling. Together, all these things require us to think about user-centered design when dealing with extended reality systems. So, how does this overall data flow work? We have already briefly mentioned that users' actions and positions are tracked and taken as input.

Input data is processed and the environment is rendered accordingly. The XR engine, that is the middleware, utilizes the hardware resources to generate output, and the output is supposed to excite the user's senses using the output devices. And here, input modeling and output or perception modeling are important considerations from the user's point of view so that the user is able to provide input easily and able to understand output easily. So, as the schematic shows, XR systems are highly interactive like any other interactive systems; consideration of human factors is very important to understand how easily a user can provide input and to understand how easily a user can perceive the output. So, we are going to get into more details of the human factors, particularly how those extended reality, or rather, perceiving extended reality or interacting with extended reality systems. So, in the next few lectures, we are going to devote some time to the discussion of human factors relevant to extended reality systems. For further details on interactive system design and implementation, you can refer to this book.



human-computer interaction as well as another NPTEL MOOC course that I have developed earlier on design and implementation of human-computer interfaces. So, most parts of this lecture are taken from these two sources. So, you can refer to those for more details. Here we will not go into any further details of interactive system design and development as that is out of scope. So, you may go through either of these sources or both to learn in more detail about this topic.

So, with that, we would like to conclude this topic. In summary, we can say that we have talked about interactive systems, the basic idea and how they are different from any other system, what the design concerns of interactive systems are, and also why we should bother about interactive systems, and how XR is related to interactive system development. Next, we are going to talk about, as I have already mentioned, human factors that are relevant for extended reality systems. So, looking forward to meeting you all in the next lecture.

Hope you have enjoyed the topic and understood the concepts. Thank you. Happy learning.